Point-by-Point Response to Reviewers

TUSCO Manuscript

December 21, 2025

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Introduction

We thank the reviewers for their thoughtful and constructive feedback. We have addressed each point and revised the manuscript accordingly. Below are our point-by-point responses.

1 Reviewer #1

Reviewer's Overall Assessment

In this manuscript, Liu and colleagues present a framework to test long-read transcriptome assembly pipelines and underlying datasets, based on a set of "simple" genes that are well-expressed and with a single isoform. These sets are carefully chosen for human and mouse, and the accompanying website is very easy to navigate. The TUSCO benchmarking framework is integrated into SQANTI3, a popular QC tool, which will make its adoption easy. **The work is logical, well supported and presented very clearly.**

Documentation is excellent.

Response

We appreciate the reviewer's positive assessment and the kind words about the documentation.

1.1 Comment 1: Distinction between TUSCO-novel and TUSCO-decoy

Reviewer Comment

The distinction between TUSCO-novel (annotation from those genes is removed) vs TUSCO-decoy (a misleading annotation is added in its place) should be made explicit, with results presented for both (along with the results when the full annotation is provided). It's debatable whether it's "fair" to provide inaccurate annotation. Minimap2 uses splice junction annotation for mapping so the alignments may already be biased before entering isoform discovery.

Response

We apologize for the confusion—there is **no separate "TUSCO-decoy" approach**. What we previously referred to as a "decoy" is part of TUSCO-novel: it removes the true transcript from the annotation and replaces it with an artificial variant. To avoid ambiguity, we have removed the term "decoy" from the revised manuscript (abstract, line 67).

In TUSCO-novel, for each single-isoform gene in the TUSCO gene set, we generate an artificial transcript with modified splice junctions using canonical motifs (GT/AG) found in the genome sequence. This artificial transcript replaces the true annotation entirely. Tools must then reconstruct the correct (but now unannotated) isoform from the sequencing reads despite the presence of this misleading reference. See the Methods (lines 153–155) for technical details. We chose to replace rather than simply delete annotations because gene-level annotations are largely complete in human and mouse, while isoform-level annotations are often incomplete. This design mirrors a common real-world setting: the gene is known, but the splice pattern may be novel (e.g., tissue-, disease-, or condition-specific isoforms where annotations are unreliable). Simply deleting the annotation would not test this capability.

Fig. 4b presents results for both conditions in a dumbbell-plot format, directly comparing each metric under the correct reference annotation versus under TUSCO-novel. A quantitative summary is provided in Supplementary Table 2.

Regarding minimap2 splice junction bias, we share this concern. To prevent annotation bias from affecting our evaluation, *we deliberately excluded all reference annotations during the minimap2*

alignment stage. Reads were aligned using only the genome sequence (FASTA), not gene models, ensuring alignments reflect genuine sequence evidence rather than annotation guidance. This design choice, while reducing initial alignment sensitivity compared to typical workflows, ensures that TUSCO-novel evaluates end-to-end, data-driven isoform discovery—the capability needed when working with incomplete or misleading annotations.

Changes in Manuscript: We have updated the abstract to clarify the TUSCO-novel approach and removed the term "decoy" to avoid confusion:

> "Masking transcripts in the TUSCO gene set and replacing them with modified splice variants in the annotation creates a TUSCO-novel challenge that assesses reconstruction of the true, now-unannotated isoforms."

1.2 Comment 2: Types of isoform transcriptome reconstruction

> **Reviewer Comment**
>
> Please make explicit what types of isoform transcriptome reconstruction can be assessed with TUSCO: genome-guided de novo (no annotation provided), genome-guided discovery (annotation present, look for new transcripts), genome reference-free (no genome or transcriptome prior information)?

Response

The TUSCO benchmarking framework is agnostic to the strategy adopted by the transcript reconstruction algorithm and could, in principle, support TUSCO evaluation of all three reconstruction types. However, the TUSCO gene set is presently only defined for human and mouse, both organisms with high-quality genomes and annotations. Hence, TUSCO evaluation is currently applicable only when tested on human and/or mouse data. Future developments will seek to expand the TUSCO gene set to other species.

Changes in Manuscript: We have added a sentence in Section 2.2 ("TUSCO benchmarking framework in SQANTI3") to explicitly state this capability:

> "Note that the TUSCO benchmarking framework is agnostic to the reconstruction strategy used by a transcript reconstruction algorithm. In principle, TUSCO evaluation can assess all three major classes of transcriptome reconstruction tools: genome-guided de novo (no annotation provided), genome-guided discovery (annotation provided, with the goal of identifying novel transcripts), and genome reference-free (no genome or transcriptome prior information). TUSCO evaluation can assess any such method as long as it is applied to human or mouse data, the two species for which the TUSCO gene set is currently defined."

1.3 Comment 3: Length biases in long-read sequencing protocols

> **Reviewer Comment**
>
> All the long-read sequencing protocols have strong length biases (PacBio against short or long transcripts, ONT cDNA against even moderately long transcripts, ONT RNA against longest,...); with log scales it's hard to figure out the relevant transcript lengths in relation to these biases. Consider binning into <600 bp, 600-1kb, 1kb-2kb, . . . 4kb+ or similar.

Response

We agree that log scales can obscure specific length biases relevant to long-read protocols. We have revised Fig. 2c and Supplementary Fig. 1 panels a–d to display transcript length distributions in discrete bins (<600 bp, 600–1kb, 1kb–2kb, 2kb–4kb, >4kb) as suggested. This visualization more clearly demonstrates how transcripts in the TUSCO gene set span the relevant length categories compared to GENCODE annotations and spike-in controls.

Changes in Manuscript: We have updated Fig. 2c and Supplementary Fig. 1 panels a–d and their captions to present the transcript length distributions in the suggested bins instead of the log10 scale.

1.4 Comment 4: Very deep datasets

> **Reviewer Comment**
>
> Would very deep datasets (100M reads?) cause issues, because rare transcription noise would still pass the threshold as a new isoform?

Response

We agree that very deep sequencing (e.g., 100M reads) captures more rare events, but whether this "noise" is reported as novel isoforms depends heavily on the filtering strategies of the reconstruction tools:

- **IsoQuant (Relative Thresholds):** IsoQuant filters novel transcripts that do not exceed 2% of the maximum graph coverage. As sequencing depth increases, the absolute number of reads required to meet this 2% threshold rises proportionally. Since stochastic transcription noise typically remains at low absolute frequencies, it fails to cross this dynamic threshold even in deep datasets.
- **Bambu (Probabilistic Modeling):** Bambu groups reads into "Read Classes" based on shared splice patterns and employs probabilistic modeling to identify high-confidence isoforms. Rare, stochastic noise that does not form consistent, high-confidence read classes is effectively filtered out during this consolidation process.
- **Iso-Seq + SQANTI3 (ML Filtering):** The SQANTI3 Machine Learning filter uses a Random Forest classifier trained on structural and quality attributes (e.g., TSS/TTS support, canonical junctions) to distinguish true isoforms from artifacts. This classifier

learns from the features of annotated transcripts, which include expression levels, and it may call a rare transcript as true depending on how similar their features are to those of the annotated transcripts.

If a tool fails to filter out this transcriptional noise and reports it as an isoform, TUSCO evaluation will identify it as a False Positive (FP), as genes in the TUSCO gene set are strictly single-isoform. Thus, TUSCO evaluation remains a valid benchmark for assessing a tool's ability to control for noise in deep datasets.

1.5 Comment 5: Example TUSCO report

> **Reviewer Comment**
>
> Consider providing an example report of TUSCO.

Response

We now include an example HTML report generated by the TUSCO benchmarking framework. It illustrates the diagnostic plots and QC metrics the framework produces, and shows how users can assess long-read data quality and reconstruction performance. The example TUSCO report is available at: `https://github.com/ConesaLab/SQANTI3/example/tusco_output/tusco_report.html`

1.6 Comment 6: Universal set download on website

> **Reviewer Comment**
>
> On the TUSCO website, it's easy to download the tissue-specific gene sets but I couldn't see how to download the universal set. It can be found on the SQANTI3 Github but it would be convenient to put it on the TUSCO website too.

Response

We have updated the TUSCO website (tusco.uv.es) to include a direct download link for the universal TUSCO gene set, alongside the tissue-specific sets.

2 Reviewer #2

Reviewer's Overall Assessment

The authors report on the TUSCO benchmarking framework, a newly created benchmark for evaluating the accuracy of transcript reconstruction with long RNA sequencing reads (LR). The TUSCO gene set consists of transcripts encoded in the human or mouse genome and therefore captures both technical and biological aspects that are not captured by artificial spike-in datasets (Sequins, SIRVs) or simulation models, such as RNA degradation and extraction artifacts.

The manuscript is well-written and substantiated, with in-depth evaluations using the TUSCO benchmarking framework that demonstrate its advantages. The construction of the 'novel variant' test set is ingenious.

The primary drawback of the TUSCO benchmarking framework is that it cannot be used to assess transcript reconstruction in the presence of multiple isoforms per gene, which is one of the main promises and goals of long read RNA sequencing and LR transcriptomic analyses. The authors already recognize this as a limitation and address it in their Discussion section.

Nevertheless, overall, **the TUSCO benchmarking framework is the first endogenous benchmark that is organically integrated with the experiment, and the only one to reflect the complexities in LR RNA datasets resulting from RNA degradation and RNA capture. Therefore, it represents an original and valuable complement and addition to the benchmark sets for LR RNA sequencing.**

Code availability: While I did not assess the current SQANTI3 code, I was able to locate the general (cross-tissue) datasets and access the web site with the tissue-specific transcripts based on the instructions and links provided.

Response

We appreciate the reviewer's positive assessment of the TUSCO benchmarking framework and the note on code/data availability; we are glad the datasets and website were easy to find.

2.1 Comment 1: Statistical testing for SIRVs vs TUSCO

Reviewer Comment

Section 2.4 and Fig. 3b: When comparing the performance metrics for TUSCO and SIRVs, the authors mention that no significant differences were found for the TP and FN categories, whereas the figures suggest otherwise, i.e., SIRVs overtaking TUSCO for TPs. This could be because the statistical test was formulated for 'TUSCO > SIRVs', for which the null hypothesis could not be rejected. The authors should incorporate testing for 'SIRVs > TUSCO'.

Response

Thank you for pointing this out. We revised this analysis to explicitly test both directions. Concretely, for Fig. 3b we now use a two-sided paired sign-flip permutation test on the paired pipeline-level differences ($n = 6$ paired pipelines per species), which is exact for small n and avoids distributional assumptions.

Following the logic of Fig. 3b, we first asked why one would compare TUSCO (endogenous controls) to spike-ins (SIRVs/Sequins): whether endogenous controls behave differently across pipelines/species than synthetic spike-ins. The primary observation is that spike-ins show higher TP rates than TUSCO evaluation (human TP $p = 0.046$; mouse TP $p = 0.015$). Decomposing where the "missing" TP goes shows it is driven by increases in PTP and FN rather than FP: PTP and FN are higher under TUSCO evaluation (human PTP $p = 0.046$, FN $p = 0.046$; mouse PTP $p = 0.046$, FN $p = 0.046$), while FP are not elevated (human FP $p = 0.077$; mouse FP $p = 0.877$). Intuitively, PTP indicates partial reconstruction (often truncation/incomplete capture) and FN indicates missed transcripts (often depth/protocol/RNA-quality limitations).

Our mechanistic interpretation is that, because TUSCO transcripts are endogenous, they are exposed to RNA quality, extraction and library biases, and depth limitations that intact spike-ins do not experience. We test this directly in Fig. 3c: the fraction of fully reconstructed TUSCO transcripts ($\frac{TP}{TP+PTP}$) correlates strongly with RIN, whereas Sequins does not. As secondary validation, we manually inspected representative TUSCO false positives and found technical artifacts unsupported by long- and short-read evidence (Supplementary Notes 1–4), and the TUSCO gene set is defined from large short-read compendia as widely expressed, supporting the interpretation of FN as genuine non-recovery rather than lack of expression.

Changes in Manuscript: We have rewritten the corresponding paragraph to reflect this logic and report two-sided paired test results:

> "Across both species, spike-ins showed higher TP rates than TUSCO evaluation (two-sided paired sign-flip permutation test across six pipelines, $n = 6$ pairs; human TP $p = 0.046$, mouse TP $p = 0.015$). Decomposing this TP loss (still Fig. 3b) shows that it is driven by increases in PTP and FN rather than an explosion of FP: PTP and FN are higher under TUSCO evaluation (human PTP $p = 0.046$, FN $p = 0.046$; mouse PTP $p = 0.046$, FN $p = 0.046$), whereas FP are not elevated (human FP $p = 0.077$, mouse FP $p = 0.877$). Intuitively, PTP reflects partial reconstruction (often truncation or incomplete capture), whereas FN reflects missed transcripts (often due to depth, protocol, or RNA quality limitations)."

2.2 Comment 2: Reference annotations for novel transcript evaluation

> **Reviewer Comment**
>
> In evaluating the programs' (Bambu, StringTie2, FLAIR, Iso-Seq+SQANTI3 ML) performance for 'novel' transcripts, it is not clear what reference annotations were provided to the programs to create the radar plots. More specifically, did the annotations include all genes and variants in GENCODE, or only those related to the TUSCO gene set? Also, how were transcripts predicted by the programs at loci outside of the TUSCO set accounted for?

Response

We appreciate the reviewer's request for clarification regarding the reference annotations used in the 'novel' transcript evaluation.

1. Reference Annotations Provided: For the 'novel' transcript benchmarking (Fig. 4), the reconstruction programs (Bambu, StringTie2, FLAIR, IsoQuant) were provided with a **comprehensive reference annotation** that included **all genes and isoforms in GENCODE**, with a specific modification restricted to the TUSCO loci. For the TUSCO genes, the true biological transcript models were removed and replaced by creating an entirely different splice pattern with artificial donor and acceptor sites at all junctions, supplying this artificially modified transcript as the sole reference. This modified full-genome annotation (e.g., `tusco_human_full.gtf`) served as the input guide for the genome-guided tools. For the Iso-Seq pipeline (official PacBio Iso-Seq pipeline + SQANTI3-ML filtering with complementary short-read sequencing data), this modified annotation was used during the SQANTI3 QC and Machine Learning filtering steps.
2. Accounting for Transcripts Outside TUSCO Loci: Transcripts predicted by the programs at loci outside of the TUSCO gene set were **excluded from the performance metrics calculation**. As detailed in the "Methods" section under "TUSCO and SIRV metrics calculation", our evaluation was strictly restricted to reconstructed transcripts that mapped to TUSCO genes, as these are our "ground truth" (identified by matching gene symbols or IDs in the SQANTI3 classification). Consequently, predictions at other genomic loci did not influence the sensitivity, precision, or other metrics shown in Fig. 4.

2.3 Comment 3: FLAIR collapse step

> **Reviewer Comment**
>
> Section 4.3, Transcriptome reconstruction with FLAIR. Was the 'collapse' step run with or without a reference annotation?

Response

The FLAIR "collapse" step was run **with a reference annotation** provided. As described in Section 4.3 (Transcriptome Reconstruction with FLAIR), alignment files were refined with the FLAIR `correct` subcommand to adjust splice junctions and transcript boundaries based on an external reference GTF annotation. This approach ensures that transcript boundaries are guided by known annotations while still allowing for novel isoform discovery based on read evidence.

Changes in Manuscript: We have clarified this in the Methods section to make the annotation usage explicit.

3 Reviewer #3

Reviewer's Overall Assessment

To improve the benchmarking of long read sequencing (LRS) methods, the authors present the TUSCO benchmarking framework (Transcriptome Universal Single-isoform Control). The TUSCO benchmarking framework includes a computational pipeline applied to varied RNA-seq data which selects endogenous, widely-expressed, single-isoform genes to serve as internal controls.

The TUSCO manuscript presents important ideas which are worthy of publication. The strategy of defining internal controls via expressed single-isoform genes is compelling, and the authors make good arguments for the utility of this approach. However, the current manuscript suffers from major gaps that should be addressed prior to publication.

Response

We appreciate the reviewer's positive assessment and the constructive feedback.

3.1 Comment 1: Define benchmarking goals

Reviewer Comment

At a high-level, what is this work about? The authors state it is about "benchmarking long reads transcriptomics", though what exactly is involved in such benchmarking is never defined. While this is implicitly addressed later by the stats the authors derive, the goals of benchmarking should be clearly stated/defined.

Response

Thank you for this suggestion. The goal of the TUSCO benchmarking framework is to evaluate the accuracy and completeness of long-read transcript reconstruction by measuring:

1. **Sensitivity (detection completeness):** Whether transcripts known to be expressed are successfully detected by the reconstruction pipeline.
2. **Precision (structural accuracy):** Whether detected transcripts match the expected reference structure without spurious splice sites or artificial variants.
3. **Robustness to sample quality:** How well pipelines perform under real-world conditions including RNA degradation, sequencing depth variations, and library preparation artifacts.
4. **Novel isoform discovery capacity:** The ability to detect true transcripts even when reference annotations are incomplete or misleading.

Changes in Manuscript: We have added an explicit statement of benchmarking goals in Section 2.2 (TUSCO-SQANTI3: A Unified Framework for Transcript Benchmarking):

> "The TUSCO benchmarking framework aims to evaluate four key aspects of long-read transcript reconstruction: (1) sensitivity—whether expressed transcripts are detected; (2) precision—whether detected transcripts match expected structures; (3)

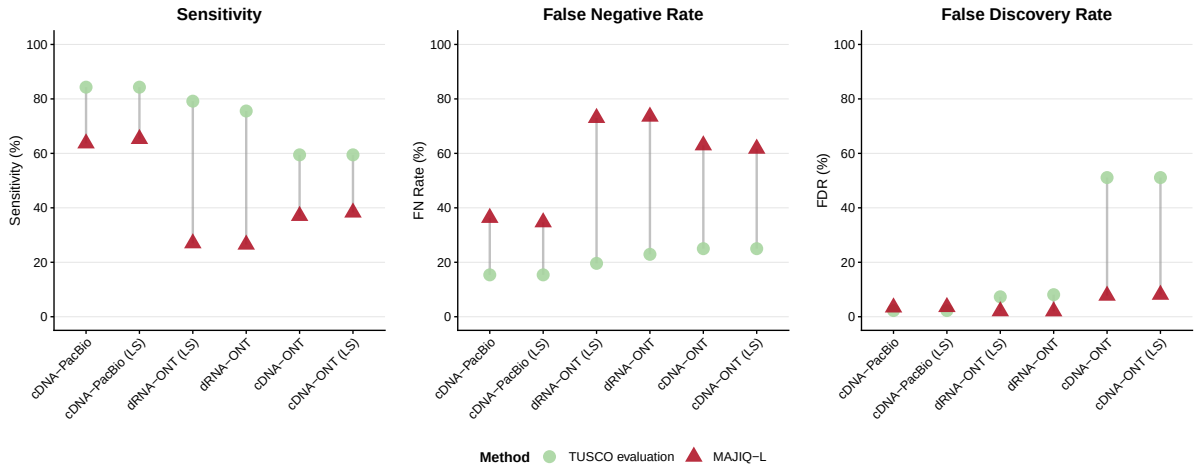
robustness—performance under real-world sample quality variations; and (4) novel discovery capacity—ability to identify true transcripts with incomplete annotations."

3.2 Major Gap 2: Comparison with short-read baseline approach

> **Reviewer Comment**
>
> In the abstract the authors postulate benchmarking has been done so far by two approaches: Spiked-in transcripts and synthetic data generation. The entire manuscript is focused on comparing to spiked-in transcripts to show TUSCO advantage. However, these are definitely not the only approaches. Specifically, recent work (MAJIQ-L, Han et al, Genome Research 2024) used short-reads sequenced from the same samples as a baseline to compare different long reads algorithms and long reads technology (ONT vs PacBio). While on the surface these are different evaluations (full isoforms reconstruction is not possible with short reads) in practice much of what the authors aim to assess (and what users care about) is totally achievable by analyzing short read sequences. Specifically, while transcript start/end are not available directly from short reads, Han et al pipeline allows users to assess where long reads report junctions, splice sites, and intron retention that do not exist by short reads (or annotation). Clearly, there could be advantages and disadvantages to assessing long reads with TUSCO vs. using short reads. For example, short reads require the matching experiment to be done with short reads and is not suitable for assessing start/end of transcripts. It also uses short-reads as the baseline which, by definition, cannot be considered an absolute truth but more of an approximation with its own biases. Nonetheless, using short read carry important potential benefits: It can be done for any condition easily and will not be sensitive to the definition of synthetically introduced TUSCO transcripts or to actual splice variants occurring in TUSCO genes in a condition/disease. It also gives a much broader "natural" set of genes with all their built in complexity without resorting to special case genes which may be biased in some unknown way (more on that later). It also gives a wider spread of TPM, gene length etc. to assess performance, important factors glossed over in the current manuscript. Consequently, it may offer much more accurate assessment of false negatives (see below).
> While the above qualitative assessment may be sensible it is also speculative and not quantitative. Given the importance of this to improving long reads transcriptomics benchmarking (the stated goal of this manuscript), the authors must investigate this thoroughly. Indeed Han et al used much of the same data from the long reads consortium and all their analysis/results are available. Thus, it should be relatively straightforward to compare TUSCO on the same data. The fact that Han et al used this orthogonal approach/method (MAJIQ based analysis) with the same data can help shed more light on TUSCO.

Response

We thank the reviewer for this suggestion. We have performed a systematic comparison of the benchmarking metrics that can be computed with TUSCO evaluation versus MAJIQ-L, and present the results below (Supplementary Fig. 3).



Supplementary Fig. 3. Comparison of TUSCO evaluation and MAJIQ-L benchmarking results. Left panel: Sensitivity (TP / (TP + FN)). Middle panel: False Negative rate (FN / (TP + FN)). Right panel: False Discovery Rate ($100 \times (N - TP)/N$). Green circles represent TUSCO evaluation metrics; red triangles represent MAJIQ-L metrics. Lines connect the two methods for each pipeline. Only cDNA-PacBio consistently ranks best in both methods; dRNA-ONT and cDNA-ONT show divergent rankings that reflect different quality dimensions (RNA integrity vs. sequencing depth).

Metric Definitions

To facilitate a fair comparison, we first define how benchmarking metrics are computed with each approach. For MAJIQ-L, junctions are classified as **known** (present in the annotation) or **novel** (not in the annotation):

Metric	**Known Junctions**	**Novel Junctions**
Ground truth	All junctions identified by MAJIQ	All novel junctions identified by MAJIQ
TP	TP-known: LR known + MAJIQ	TP-novel: LR novel + MAJIQ novel
FP	FP-known: LR known, no MAJIQ	FP-novel: LR novel, no MAJIQ
TN	*Not possible (unknown expression)*	*Not possible by MAJIQ*
FN	FN-known: No LR, Yes MAJIQ	FN-novel: No LR, Yes MAJIQ novel

Note that **TN cannot be computed** with MAJIQ-L because we do not know if a gene is expressed.

False Discovery Rate (FDR) is computed as: FDR = $100 \times (N - TP)/N$, where N is the number of observed transcripts (TUSCO evaluation) or LR junctions (MAJIQ-L). For TUSCO evaluation, N = TP + PTP + FP (observed transcripts); for MAJIQ-L, N = total_LR_junctions and TP includes both known and novel.

MAJIQ-L Benchmarking Results

We applied the MAJIQ-L framework to the same LRGASP WTC11 datasets. Table 1 shows the junction-level metrics aggregated across all expressed genes:

Table 1: MAJIQ-L junction-level benchmarking results.

Pipeline	TP-known	FP-known	FN-known	TP-novel	FP-novel	FN-novel
cDNA-ONT	80,021	4,397	124,611	839	2,385	12,847
cDNA-ONT (LS)	81,343	4,447	123,289	2,221	2,904	11,465
cDNA-PacBio	138,081	3,617	66,551	939	1,312	12,747
cDNA-PacBio (LS)	139,903	3,637	64,729	2,655	1,645	11,031
dRNA-ONT	57,801	971	146,831	52	208	13,634
dRNA-ONT (LS)	58,718	980	145,914	181	237	13,505

From these values, we compute Sensitivity as (TP-known + TP-novel) / (TP-known + TP-novel + FN-known + FN-novel), FN Rate as (FN-known + FN-novel) / (TP-known + TP-novel + FN-known + FN-novel), and FDR as $100 \times (N - TP)/N$ where N is the total number of LR junctions detected.

Comparison of Benchmarking Insights

Supplementary Fig. 3 compares Sensitivity, FN rate, and FDR between TUSCO evaluation and MAJIQ-L across all pipelines. Only cDNA-PacBio consistently ranks best in both methods. The rankings for cDNA-ONT and dRNA-ONT differ substantially between TUSCO evaluation and MAJIQ-L, revealing that each method captures different aspects of data quality.

TUSCO evaluation is sensitive to sample quality, particularly RNA integrity. The cDNA-ONT pipeline shows 38.3% PTP transcripts, indicating incomplete RNA molecules with poor TSS/TTS accuracy. This results in low sensitivity (59.5%) and high FDR (51.1%). In contrast, dRNA-ONT achieves high sensitivity (75.6%) despite having only 760K reads (compared to 16.6M for cDNA-ONT), suggesting that the RNA molecules in this preparation are more intact.

MAJIQ-L, on the other hand, reflects sequencing depth because it evaluates all splice junctions rather than focusing on consistently expressed genes like those in the TUSCO gene set. The dRNA-ONT pipeline has only 760K reads (22-fold fewer than cDNA-ONT's 16.6M reads) and consequently misses many lowly-expressed junctions, leading to a high FN rate (73.5%) in MAJIQ-L. Importantly, the low FDR in dRNA-ONT (2.0% by MAJIQ-L, 8.1% by TUSCO evaluation) confirms that sample quality is good—the high FN rate is due to depth limitations, not poor sample quality. The cDNA-ONT pipeline has 16.6M reads, detecting more junctions and achieving higher MAJIQ-L sensitivity (37.0%) compared to dRNA-ONT (26.5%).

These complementary patterns demonstrate that TUSCO evaluation and MAJIQ-L assess fundamentally different quality dimensions.

The two methods differ fundamentally in their approach. MAJIQ-L evaluates junction-level metrics and can assess novel junctions not present in the annotation, while TUSCO evaluation evaluates transcript-level metrics including full transcript structure and is limited to consistently expressed genes in the TUSCO gene set. A key advantage of TUSCO evaluation is its ability to evaluate TSS and TTS accuracy through the PTP (partially true positive) category, which MAJIQ-L cannot assess. Furthermore, TUSCO evaluation provides a true ground truth based

on the reference annotation, whereas MAJIQ-L uses short-read data as an approximation of the truth.

Conclusion

TUSCO evaluation and MAJIQ-L are complementary benchmarking approaches that capture different aspects of long-read transcriptomics performance. TUSCO evaluation provides a true annotation-based ground truth rather than relying on short-read approximations, can evaluate both PTP and FN at the transcript level, requires no additional sequencing costs (no matched short-read data needed), and uniquely captures TSS and TTS accuracy through the PTP category. We acknowledge that the TUSCO gene set is not defined for novel species and has not been tested in disease settings.

3.3 Major Gap 3: False negatives in disease studies

> **Reviewer Comment**
>
> The claim that TUSCO evaluation can detect false negatives even in unusual biological conditions like disease studies is suspect. While the TUSCO control genes were selected based on an analysis of many cell types and other factors, transcript expression is known to vary widely in diverse conditions, especially disease. This point should be reframed or better argued in the manuscript.

Response

We appreciate this important concern, and we agree that transcript expression can vary substantially in disease contexts. Some genes in the TUSCO gene set may be downregulated or absent in specific diseases, potentially inflating apparent FN rates. At the same time, our selection pipeline was designed to minimize alternative splicing evidence in the TUSCO set, so we expect false alternative transcript events to remain rare.

Still, TUSCO evaluation can be useful in disease studies for:

- **Accuracy on control samples:** Many disease studies include healthy controls, which remain a good testbed for method benchmarking.
- **Sample quality indicators:** The PTP/TP ratio provides a measure of RNA integrity and degradation that applies regardless of disease state.
- **Pipeline performance baseline:** Even in disease samples, TUSCO evaluation provides a baseline for comparing different reconstruction pipelines applied to the same data.

Changes in Manuscript: We have revised the Discussion to temper claims about disease applications and added a note recommending that users interpret TUSCO evaluation results in unusual biological conditions (such as disease) with appropriate caution, using TUSCO evaluation primarily for relative pipeline comparison rather than absolute performance assessment.

3.4 Major Gap 4: Statistics interpretation

> **Reviewer Comment**
>
> Are the statistics used to assess performance supposed to represent "absolute truth" of some kind? Relative ranking of methods/algorithms by specific metrics? This is unclear and definitions/statements need to be made and supported accordingly. Clearly there are major effects of coverage, transcript length (see also point below) and the variance observed in the reported statistics is so high it leaves the reader wondering what does this all mean. For example, PTP average in mouse (p. 8 top) is 54% with 43% std (!).

Response

TUSCO metrics serve two complementary purposes:

1. **Relative ranking of methods/pipelines:** For a given dataset, TUSCO evaluation enables comparison of different reconstruction tools (e.g., Bambu vs. FLAIR vs. Iso-Seq) by providing consistent metrics across all tools. This ranking is the primary intended use.
2. **Sample quality assessment:** Within a single pipeline, TUSCO metrics (particularly PTP rates and TP/PTP ratios) indicate sample-specific factors like RNA integrity, which can vary substantially.

The high variance observed (e.g., PTP average ∼54% with 43% std in mouse) reflects *real biological and technical heterogeneity* across samples, which is precisely what TUSCO evaluation is designed to capture. This variance arises from:

- Differences in RNA integrity (RIN) across samples (demonstrated in Fig. 3c)
- Sequencing depth variations (demonstrated in Supplementary Fig. 4)
- Library preparation method differences (ONT vs. PacBio, dRNA vs. cDNA)
- Biological variation in expression levels

We agree the variance can be large, but it is expected—it reflects the sensitivity of TUSCO evaluation to real experimental factors. By contrast, SIRVs show artificially low variance (Supplementary Fig. 7) precisely because they bypass sample preparation variability.

Changes in Manuscript: We have added clarifying text in Section 2.3 explaining that TUSCO metrics should be interpreted as: (1) relative performance rankings within a dataset, and (2) indicators of sample-specific quality, with expected variance reflecting real technical heterogeneity.

3.5 Major Gap 5: Radar plots criticism

> **Reviewer Comment**
>
> Radar polygons and radar area are used throughout the manuscript for visualization and tool comparison. While these may be visually pleasing we see this as a case of form over function. Radar area is not an appropriate statistic to summarize and compare method performance (what does it actually mean?). Worse, the calculated area value depends strongly on the clockwise order of statistics; for example, for the six metrics shown in Fig. 3a, hypothetical scores ordered 1, 1, 1, 0, 0, 0 (nonzero area) have a larger radar area than 1, 0, 1, 0, 1, 0 (zero area).
>
> One wonders whether conclusions would change with a different ordering. Even in human perceptions there may be additional biases (e.g. things on the top vs on the sides). Radar polygon plots are not an intuitive way to compare tool performance because it is virtually impossible to visually compare each individual statistic across tools. The manuscript should use standard visualization methods (e.g. bar charts) allowing for comparison between statistics. It's simply hard to assess the authors' claims and converge to conclusions with these plots.

Response

Changes in Manuscript: We agree with the reviewer's critique regarding the limitations of radar plots. We have replaced all radar plots in the manuscript with dumbbell plots (Figures 3, 4, and 5) and removed radar-area-based summaries. This visualization enables direct comparison of each statistic across tools and conditions, independent of metric ordering.

3.6 Major Gap 6: TUSCO transcript length enrichment

> **Reviewer Comment**
>
> Section 2.1 makes the claim that TUSCO transcripts are comparable in length to RefSeq transcripts (Fig. 1c), but this is not actually supported by Fig. 1c, which illustrates an apparent enrichment of longer transcripts in TUSCO (even more-so given the log scale in Fig. 1c). We can envision a different manuscript showing exactly the same plots as in Fig. 1c,d and making the *apposite* claim i.e. that they are significantly different, attaching a p-value to that claim...
>
> What are the implications of using unusually long transcripts as the internal controls? Moreover, this fact is a bit surprising and should be explained, as one would not necessarily expect single-isoform genes to generate relatively long transcripts. This points to a needed discussion about what biases are introduced by using such strongly-curated transcripts as the internal controls. Coming back to points raised above

Response

The reviewer correctly observes that transcripts in the TUSCO gene set tend toward the longer end of the distribution. We have quantified this more precisely in the revised manuscript.

Why genes in the TUSCO gene set tend to be longer: Genes in the TUSCO set are enriched for single-isoform loci with highly stable annotations and broad expression. Many encode core cellular machinery—such as mitochondrial translation factors, RNA-processing components, protein trafficking regulators, and metabolic enzymes—which often require larger, multi-domain proteins and therefore longer transcripts. Because these genes are dosage-sensitive and function-

ally constrained, they are under strong selective pressure to maintain a single, canonical transcript structure, leading to reduced tolerance for alternative splicing and, consequently, greater isoform simplicity.

Implications for benchmarking:

- **Potential advantage:** Longer transcripts are more challenging for full-length reconstruction, making TUSCO evaluation a stringent benchmark that tests the upper bounds of pipeline performance.
- **Potential limitation:** Performance on short transcripts (<600 bp) may be underrepresented in TUSCO evaluation.

Changes in Manuscript: We have:

1. Revised Fig. 1c to use binned length distributions (as suggested by Reviewer 1) for clearer interpretation.
2. Added explicit acknowledgment in the Discussion that the length distribution of transcripts in the TUSCO gene set skews toward longer transcripts, which should be considered when interpreting benchmarking results.

3.7 Additional Improvement 1: TUSCO terminology consistency

> **Reviewer Comment**
>
> Throughout the manuscript, TUSCO is confusingly referred to as a set of genes, a framework, and a collection of analyses. To this reviewer, TUSCO is certainly more than a set of genes since it includes the upstream computational pipelines and analysis pipelines, which are significant contributions. We recommend using terms more specifically to refer to the full collection of pipelines and results vs. specific parts like the cross-tissue filtered gene sets.

Response

Clarification of terminology: We thank the reviewer for this suggestion. We have standardized terminology throughout the manuscript:

- **"TUSCO gene set"**: The curated collection of single-isoform genes used as ground truth.
- **"TUSCO benchmarking framework"**: The complete system including gene selection pipeline, SQANTI3 integration, and metrics calculation.
- **"TUSCO evaluation"**: The process of applying the TUSCO benchmarking framework to assess a transcriptome reconstruction.
- **"TUSCO metrics"**: The specific performance measurements (TP, PTP, FP, FN rates, sensitivity, precision, etc.).
- **"TUSCO-novel"**: The evaluation mode using modified annotations.

Changes in Manuscript: We have revised the manuscript to consistently use these terms and avoid using "TUSCO" without a qualifier when referring to a specific component (gene set, benchmarking framework, evaluation, or metrics).

3.8 Additional Improvement 2: "Biologically plausible" definition

> **Reviewer Comment**
>
> The TUSCO-novel claim of "biologically plausible isoforms" (section 2.5) is interesting, but the validity of "biologically plausible" is unclear. Were the made-up splice sites chosen to e.g. preserve splice motifs / enhancers, etc? What about exon length and GC content bias?

Response

In the current TUSCO-novel simulator implementation, the artificial isoforms are designed to be *structurally plausible* in the limited sense of satisfying basic splice-junction sequence constraints. For multi-exon transcripts, each annotated intron is modified by shifting the donor and acceptor coordinates *within the original intron bounds* while searching the gene's genomic sequence for canonical dinucleotides: `GT` at the donor and `AG` at the acceptor on the plus strand, and the corresponding reference-strand motifs `AC` (donor) and `CT` (acceptor) on the minus strand. The search is directional (donor downstream / acceptor upstream on the plus strand) with a default window of 100 bp, falling back to scanning the full intron if needed, and it enforces a minimum shift of 5 bp when a suitable motif can be found. For single-exon transcripts, the simulator introduces one synthetic intron by selecting two positions separated by at least 50 bp and then adjusting each boundary to a nearby canonical motif within 20 bp when possible.
That said, several aspects of *biological* plausibility are not modeled or guaranteed. Splice enhancer/silencer sequences are not preserved, branch point / polypyrimidine tract features are not verified, and GC content or exon-length distributions are not matched. The goal is not to generate transcripts that would necessarily be spliced in vivo, but to create misleading annotations that are superficially consistent with simple splice-site rules and could plausibly appear in a reference annotation. This tests whether reconstruction tools primarily follow the supplied annotation or instead use read evidence to recover the true splice pattern.

Changes in Manuscript: We have revised the term to "structurally plausible" rather than "biologically plausible" and clarified in the Methods that artificial splice sites preserve canonical motifs but do not guarantee functional splicing.

3.9 Additional Improvement 3: Parameter selection justification

> **Reviewer Comment**
>
> It's not clear how certain parameter values were selected. Moreover, the sensitivity of results to those choices is not clear. This could be better argued/explained. For example:
>
> - **3.a.** How were the window sizes for TSS, TTS, and CAGE decided, and what is the sensitivity of the results to those parameters?

- **3.b.** TUSCO genes are medium to highly expressed across tissues (Section 2.1); is there value in lowering the minimum expression threshold to capture lower-yet-consistently-expressed genes, and how are LRS method evaluation results affected?
- **3.c.** Splice evidence can require as little as 1 supporting read for a junction in TUSCO gene selection (Section 4.1); is there value in more stringency? Should this part of the pipeline handle known confounders such as read stacks / artifacts and require junction read support from multiple positions? How would this change the results?
- **3.d.** How were the different thresholds for mouse vs human RNAseq data decided?

Response

We conducted systematic sensitivity analyses for each major parameter category. All results are consolidated in new Supplementary Figure 8 and Supplementary Table 4.

3.a. TSS/CAGE window size validation:
The ±300 bp window was chosen to balance sensitivity against inter-gene collision. Using refTSS v4.1 on single-isoform genes, we computed two complementary metrics as a function of window size (Supplementary Fig. 8a–b). At 300 bp, the human data show 96.6% sensitivity (fraction of genes whose annotated TSS falls within the window) against only 2.1% collision (fraction with a neighboring gene's TSS within range); mouse figures are comparable (93.8% vs 4.3%). The 75th-percentile peak count at this threshold is 2, meaning roughly one quarter of genes—those with complex promoter architecture—are excluded. Because the collision rate remains below 5%, these exclusions reflect genuine alternative TSSs rather than technical artifacts from nearby genes. Narrower windows sacrifice sensitivity; wider ones admit too many alternative promoters.

3.b. Expression thresholds:
We swept four configurations spanning strict to lenient expression criteria:

Human	Prevalence	Median RPKM	Min Expr	Splice Purity	n_genes
Strict	0.99	1.0	0.10	0.90	36
Current	0.95	0.5	0.05	0.90	46
Lenient	0.75	0.2	0.02	0.90	64
More Lenient	0.55	0.1	0.01	0.90	79

Mouse	Prevalence	Median RPKM	Min Expr	Splice Purity	n_genes
Strict	0.95	1.0	0.10	0.90	30
Current	0.90	0.5	0.05	0.90	32
Lenient	0.70	0.2	0.02	0.90	40
More Lenient	0.50	0.1	0.01	0.90	58

Here, *Prevalence* is the fraction of tissues where a gene must be expressed; *Median RPKM* and *Min Expr* set expression floors; and *Splice Purity* (held at 0.90) is the fraction of tissues whose splice ratio falls below threshold.
We then benchmarked each configuration against SIRV spike-in controls across six LRGASP pipelines (Supplementary Fig. 8c–d):

Species	Config	n_genes	Median TUSCO FN%	Median SIRV FN%	Difference
Human	Strict	36	9.7%	3.7%	+6.0%
Human	Current	46	18.5%	3.7%	+14.8%
Human	Lenient	64	28.9%	3.7%	+25.2%
Human	More Lenient	79	36.1%	3.7%	+32.4%
Mouse	Strict	30	20.0%	4.5%	+15.5%
Mouse	Current	32	23.4%	4.5%	+18.9%
Mouse	Lenient	40	31.2%	4.5%	+26.7%
Mouse	More Lenient	58	49.1%	4.5%	+44.6%

TUSCO false-negative rates exceed SIRV baselines because endogenous genes are subject to RNA degradation and variable expression that synthetic spike-ins bypass. The "Current" configuration adds 10 genes (human) or 2 genes (mouse) relative to "Strict" with only a moderate rise in FN; moving further to "Lenient" nearly doubles FN rates, indicating that those additional genes are simply not reliably detected at typical sequencing depths. As demonstrated elsewhere in the manuscript, FN in the "Current" set correlates with sequencing depth, not threshold choice—lowering thresholds would include genes that remain undetected regardless.

3.c. Splice junction read support:
The "≥1 read" floor exists only to prevent divide-by-zero when $\mu = 0$; the operative filter is the relative threshold $T = \alpha\mu$, where μ is mean annotated-junction coverage and α is 0.01 (human) or 0.05 (mouse). In recount3, μ is typically large—minimum values among final multi-exon TUSCO genes are 5.1×10^5 (human) and 9.8×10^5 (mouse)—so the 1-read floor never determines inclusion. We verified this by plotting ECDFs of the ratio $\max(\text{novel})/\mu$ for TUSCO versus non-TUSCO single-isoform genes (Supplementary Fig. 8e). TUSCO genes cluster well below the chosen α (95th percentiles: 0.0075 human, 0.032 mouse), whereas excluded genes show markedly higher ratios. Read stacks and other artifacts are further mitigated by requiring junctions to be fully intragenic, strand-consistent, and ≥80 bp.

3.d. Human vs. mouse threshold differences:
Mouse and human expression distributions differ in both the number of available RNA-seq reference tracks and the typical absolute RPKM scale. Therefore, we did not enforce a single universal RPKM cutoff, but instead selected species-specific "current" expression thresholds from the same parameter sweep (Supplementary Fig. 8c) to balance gene count against a sharp rise in false negatives under more lenient settings.
Importantly, these species-specific cutoffs still target *comparably high-expression* loci: using AlphaGenome tissue-median RPKM values across all available AlphaGenome expression tracks (human: `data/hsa/alphagenome_human.json`; mouse: `data/mmu/alphagenome_mouse.json`), the median TUSCO gene falls at the 90.3545th percentile in human and the 96.5232nd percentile in mouse (Supplementary Fig. 8d). This high-expression-tail placement supports reliable detectability in benchmarking datasets, consistent with the observation that all genes in the TUSCO gene set are detected in at least one LRGASP dataset (Fig. 3d).

Changes in Manuscript: We have added Supplementary Figure 8 (parameter sensitivity analyses) and Supplementary Table 4 (parameter summary). The Methods section now provides inline justifications at the point where each threshold is first introduced, with references to these supplementary materials.

3.10 Additional Improvement 4: Cosine similarity interpretation

> **Reviewer Comment**
>
> In the TUSCO vs SIRV comparison of Section 2.3, it is not intuitive how good the concordance is from the reported cosine similarity. Is there an available negative control e.g. biased protocol or short-read-only protocol, or even "reasonable" simulated result that would illustrate a baseline cosine similarity for this high-dimensional comparison?

Response

Cosine similarity measures the angular similarity between two 6-dimensional metric vectors (Sn, nrPre, Redundancy^{-1}, 1-FDR, PDR, rPre). A value of 1.0 indicates identical relative performance profiles, while values near 0 indicate orthogonal profiles. Values above 0.99 suggest that TUSCO evaluation and SIRVs report virtually identical relative performance, indicating both benchmarks assess the same underlying pipeline quality. Moderate concordance in the 0.94–0.99 range suggests some divergence, potentially due to TUSCO evaluation's sensitivity to factors SIRVs do not capture, such as RNA degradation.
While we did not include a formal negative control such as a deliberately biased protocol, we can offer some context. The lowest observed cosine similarity (0.9493 for WTC11 cDNA ONT) still indicates strong concordance. By comparison, a randomly shuffled metric vector would yield cosine similarity of approximately 0.5–0.7 by chance, and a truly discordant benchmark evaluating different genes entirely would show values approaching 0.

Changes in Manuscript: We have added interpretation guidance for cosine similarity values in the Results section, noting that values >0.95 indicate strong concordance while values <0.90 would suggest meaningful benchmark divergence.

3.11 Additional Improvement 5: TP coverage variability

> **Reviewer Comment**
>
> The TP coverages reported in Section 2.5 are highly variable (via the +), is there more to say about when it is higher vs lower?

Response

The large "±" reflects dispersion across *individual reads* supporting transcripts labeled TP. We find that TP per-read coverage is systematically **higher and more stable for short reference transcripts**, and **lower and much more variable for long reference transcripts**. This is consistent with length-dependent truncation (RNA degradation, incomplete cDNA synthesis/-capture), which becomes more likely as transcript length increases.
Concretely, stratifying Iso-Seq TP reads by the reference transcript exonic length shows:

- **Human (WTC11 PacBio cDNA):** for TP reads mapping to transcripts ≤1 kb, mean per-read coverage is 95.7% with SD 8.3%; for 2–3 kb it is 100.2% ± 1.7%; but for longer (3–5 kb) it drops to 80.4% ± 30.5%.

- **Mouse (ES PacBio cDNA):** TP coverage remains high for most length bins (e.g., 3–5 kb: $97.1\% \pm 8.1\%$), but declines for very long transcripts (>5 kb: $88.4\% \pm 21.0\%$).

Thus, TP per-read coverage is highest when reads are near full-length (shorter transcripts and/or high-integrity molecules) and lowest when truncation is prevalent (longer transcripts and degraded/partially captured molecules), which inflates the SD even when the median coverage remains close to 1 for many bins.

Changes in Manuscript: We now explicitly describe this length dependence and report the stratified statistics to clarify when TP per-read coverage is higher vs. lower.

3.12 Additional Improvement 6: Per-read coverage definition

> **Reviewer Comment**
>
> It is not clear why per-read coverage is defined and used as in Section 2.5.

Response

Why we define and use per-read coverage here: In Section 2.5, we ask why some TUSCO gene set loci are TP (full match, including boundaries) while others are PTP (same junction chain, but shifted TSS/TTS). For Iso-Seq, many PTPs are expected from 5'/3' truncated molecules (e.g., RNA degradation or incomplete capture): reads can support the correct junction chain but fail to span the true transcript ends, yielding boundary-shifted models unless transcript ends are refined. Per-read coverage quantifies this end-completeness as a read-length ratio relative to the reference isoform.
Definition: For each supporting read, we compute $f = L_{\text{read}}/L_{\text{exonic}}$, where L_{exonic} is the total exonic length of the *true reference isoform*, and L_{read} is the observed Iso-Seq read length (PacBio cDNA; FLNC). With this definition, $f \approx 1$ indicates a near full-length molecule, $f \ll 1$ indicates truncation, and f can exceed 1 when a read is longer than the reference isoform (e.g., extra UTR/poly(A) sequence or boundary differences).

1. **TP vs. PTP boundary effects:** If PTP is driven by truncation, PTP-supporting reads should have lower f than TP-supporting reads. Consistent with this, TP reads show high coverage (median 98.3% in human), whereas PTP reads show substantially lower coverage (median 31.7% in human).

2. **Interpretation in our PacBio cDNA data:** Lower f indicates more truncated/end-incomplete molecules (consistent with degradation and/or incomplete capture), which explains boundary-shifted transcript models when TSS/TTS are not refined.

Changes in Manuscript: We have added explicit justification for this metric definition in Section 2.5 and clarified its interpretation in the figure caption.

Reviewer #3 (Remarks on code availability)

> **Reviewer Comment**
>
> *Haven't reviewed the code at this stage.*

4 Reviewer #4

Reviewer's Overall Assessment

In the manuscript by Tianyuan Liu et al. authors introduce the Transcriptome Universal Single-isoform COntrol (TUSCO), which is a curated reference set of genes lacking alternative isoforms. The aim is to provide an internal ground truth for benchmarking purposes. The TUSCO benchmarking framework evaluates precision by identifying transcripts deviating from reference annotations and assesses sensitivity by verifying detection completeness in human and mouse samples.

The study is very extensive and detailed what makes that the following the narration is very tricky, although the article is well structure and very well written. Simply the complexity of topic is high and it is difficult to provide an easy solution here.

I am confident that the TUSCO benchmarking framework will become a widely used approach for benchmarking. It has its own limitations which were also specified by the authors. The main is that it is "restricted to single-isoform loci and thus cannot evaluate transcript reconstruction accuracy within highly complex multi-isoform genes". To overcome this limitations authors introduce a set of artificial isoforms derived from genes in the TUSCO gene set. It is a nice trick which allows to assess some properties of the pipelines but still is not perfect. Authors are aware of it and are planning further work in this direction.

Response

We thank the reviewer for seeing the value of our work and for encouraging us to continue development of the TUSCO benchmarking framework.

4.1 Comment 1: Section 2.4 - Correlation interpretation

Reviewer Comment

It is stated: "A pronounced negative correlation (Pearson r $= -0.8$, p $= 1.7 \times 10^{-3}$; n $= 12$ pipelines) was observed, indicating that as sequencing coverage and read length increase, the number of false negatives decreases."

An inspection of Supplementary Fig. 3 reveals that in fact we do have a two clearly separated groups of points. One from ONT and second by PB which are strongly separated both by False Negative rate as well as composite metric. The true source of observed phenomenon here is hidden in sequencing technology differences and is not fully proven here. Thus further statements coming from this part are overinterpretation.

Response

We thank the reviewer for pointing this out. We agree that the earlier correlation framing was an overinterpretation: in this LRGASP subset, FN rates are primarily separated by sequencing platform/protocol (ONT versus PacBio).

Changes in Manuscript: We revised the text in Section 2.4 accordingly and updated Supplementary Fig. 4 to a grouped raincloud plot summarizing FN rates by platform and library type. The revised text reads:

> "Finally, we summarized FN rates across LRGASP datasets grouped by sequencing platform and library method (Supplementary Fig. 4). In this subset, FN differences are dominated by platform/protocol (PacBio cDNA lowest; ONT higher), so Supplementary Fig. 4 is presented as a platform-stratified comparison."

4.2 Comment 2: Section 2.5 - TUSCO-novel evaluation

> **Reviewer Comment**
>
> The capacity of reconstruction tools to detect unannotated isoforms is assessed with used of introduced TUSCO-novel reference set. The authors note that tools which heavily rely on annotation fail to recover new isoforms as defined in TUSCO-novel. That is true, but what was not highlighted or noted is the fact that preferred by authors Iso-Seq+SQANTI3 is not way better. It has marginal drop of performance between "reference annotation" and TUSCO-novel due to the fact that is dealing with "reference annotation" noticeably worse than other approaches. Thus ability of TUSCO-novel to evaluate novel isoforms discovery is not that remarkable I would say.

Response

The reviewer makes an astute observation. Iso-Seq + SQANTI3 ML indeed shows lower absolute performance with the reference annotation than strongly annotation-guided tools, and its minimal change under TUSCO-novel reflects this lower baseline rather than superior novel discovery. For example, summarizing Fig. 4b by a composite score defined as the mean of the six TUSCO metrics (Sn, nrPre, Redundancy^{-1}, $1 - \text{FDR}$, PDR, rPre), the median reference score is 76.29 (human) for Iso-Seq + SQANTI3 ML versus 100.00 for Bambu, while the median TUSCO-novel change for Iso-Seq + SQANTI3 ML is near zero (-0.73% in human; Supplementary Table 2).
The key insight from TUSCO-novel is not absolute performance but annotation dependence. Bambu and StringTie2 are strongly reference-guided and therefore highly dependent on reference annotation; in settings where gene-level annotation is largely complete but isoform-level annotation is missing or incorrect, this reliance can limit their ability to detect genuinely novel isoforms. Consistent with this, when the annotation at TUSCO loci is replaced by an incorrect splice pattern, Bambu and StringTie2 fail substantially to recover the true isoforms despite read support. Iso-Seq + SQANTI3 ML, by contrast, performs consistently regardless of annotation status. The lower baseline for Iso-Seq arises because the pipeline lacks TSS/TTS refinement against the reference, resulting in more PTP calls even when the junction chain is correct. This reduces TP rates but makes the pipeline inherently reference-agnostic.
The practical implication is that in scenarios where reference annotations may be incomplete or misleading—such as novel tissues, disease states, or non-model organisms—a pipeline with lower but stable performance may be preferable to one with high reference-guided performance that collapses without correct annotation.

Changes in Manuscript: We have revised Section 2.5 to explicitly acknowledge this trade-off:

> "Iso-Seq + SQANTI3 ML showed minimal performance change under TUSCO-novel (-1.9% to $+2.7\%$), though with lower absolute performance than annotation-guided tools. This stability arises from the pipeline's reference-agnostic design, which maintains consistent (if moderate) performance regardless of annotation quality—a poten-

tially valuable property when working with incomplete or unreliable references."

4.3 Comment 3: Section 2.6 - Replicate number recommendations

> **Reviewer Comment**
>
> It is stated: "... an optimal design includes at least three biological replicates (Fig. 5b; Supplementary Table 3). By contrast, analyses of SIRV spike-in controls yielded highly consistent concordance metrics across 1–5 replicates, providing little guidance on replicate number optimization (Supplementary Fig. 6)."
>
> An inspection of Fig. 5 together with Supplementary Fig. 6 reveals that there are no drastic differences across 2-5 replicates for both approaches. Thus statement that for TUSCO a good choice is 3+ while for SIRV it is hard to say is overinterpretation. Also here I do not understand what Supplementary Table 3 is adding here.

Response

We agree that the differences between 2–5 replicates are modest for both TUSCO evaluation and SIRVs, and our original phrasing may have overstated the guidance provided.
The most substantial improvement occurs between 1 and 2 replicates, with FDR dropping from 32.9% to 12.2% in brain and from 34.8% to 19.0% in kidney. Beyond 2 replicates, gains are incremental. The key difference between the two benchmarks is that TUSCO metrics do change with replicate number (albeit modestly after 2 replicates), while SIRVs are essentially invariant. This demonstrates that TUSCO evaluation captures real experimental variation that SIRVs miss, even if the absolute differences are small. Regarding Supplementary Table 3, the per-sample FLNC read counts provide the depth context needed to interpret these replicate effects and to generalize them: they allow readers to compare their expected per-sample yield to ours and, under a fixed sequencing budget, to evaluate the trade-off between sequencing deeper versus adding replicates.

Changes in Manuscript: We have revised Section 2.6 to present findings more objectively:

> "Collectively, these results indicate that the TUSCO evaluation effectively reveals the effect of replication on transcript detection accuracy. The largest precision gain occurred between one and two replicates, with additional replicates (3–5) yielding incremental improvements at this sequencing depth (Supplementary Table 3). Thus, for this experimental setup (mouse brain/kidney, PacBio cDNA, FLAIR), at least two biological replicates substantially improve precision, with diminishing returns thereafter; additional replicates can still provide modest sensitivity gains (notably in kidney) via consensus boundary refinement (Fig. 5b). By contrast, analyses of SIRV spike-in controls yielded highly consistent concordance metrics across 1–5 replicates, providing little guidance on replicate number optimization (Supplementary Fig. 7)."

5 Reviewer #5

Reviewer's Overall Assessment

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

No additional comments provided.